

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Regular/Supplementary Winter Examination – 2024****Course: S.Y. B. Tech Branch: Electronics and Computer Engineering/ Semester: III****Electronics and Computer Science:****Subject Code & Name: (BTECPC303) Programming, Data Structure and Algorithm using C****Max Marks: 60****Date: 10/02/2025****Duration: 3 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Which of the following is an example of a non-linear data structure?	(L1, CO1)	1
	a. Array b. Stack c. LinkedList d. Tree		
2	Which of the following is NOT a type of data structure?	(L2, CO1)	1
	a. Array b. Stack c. Hash Table d. Compiler		
3	Which data structure is best for implementing a LIFO (Last In, First Out) system?	(L3, CO2)	1
	a. Stack b. Queue c. Tree d. Linkedlist		
4	What is the time complexity of searching in a balanced binary search tree?	(L2, CO4)	1
	a. O(1) b. O(n) c. O(log n) d. O(n log n)		
5	In which type of linked list does the last node point to the first node?	(L1, CO1)	1
	a. Singly Linked List b. Doubly Linked List c. Circular Linked List d. None of the above		
6	What is the worst-case time complexity of linear search?	(L1, CO5)	1
	a. O(1) b. O(n) c. O(log n) d. O(n log n)		
7	What is the key difference between a stack and a queue?	(L2, CO3)	1
	a. Stack is LIFO, Queue is FIFO b. Both are LIFO c. Stack is FIFO, Queue is LIFO d. Both are FIFO		
8	Which graph traversal algorithm uses a queue?	(L3, CO4)	1
	a. Depth First Search b. Breadth First Search c. Inorder Traversal d. Postorder Traversal		
9	Which of the following operations is not performed by a binary search tree?	(L4, CO5)	1
	a. Insert b. Delete c. Sort d. Reverse		
10	What is the time complexity of bubble sort in the worst case?	(L2, CO5)	1
	a. O(1) b. O(n) c. O(n ²) d. O(n log n)		

11	Stack overflow occurs in the operations				(L3, CO2)	1
	a. Push	b. Pop	c. display	d. StackEmpty		
12	Which type of queue overcomes the limitation of a linear queue by reusing vacant spaces?				(L2, CO3)	1
	a. Simple Queue	b. Circular Queue	c. Priority Queue	d. None of the above		
Q. 2	Solve the following.					12
A)	Explain the following operations on singly linked lists a) insertion of node at beginning and middle b) Traversal				(L2, CO1)	6
B)	Write differences between array and linked list.				(L3, CO1)	6
Q. 3	Solve the following.					12
A)	Define queue? Explain enqueue and dequeue operation for simple queue?				(L2, CO3)	6
B)	Explain type of Queue? Compare Simple Queue and Circular queue? Advantages and Disadvantages of each?				(L4, CO3)	6
Q. 4	Solve Any Two of the following.					12
A)	Write an algorithm for binary search and explain its time complexity?				(L3, CO5)	6
B)	Write an algorithm for Insertion sort with an example.?				(L4, CO5)	6
C)	Write a pseudo code for Linear Search? Explain with example				(L3, CO5)	6
Q. 5	Solve Any Two of the following.					12
A)	Write an algorithm to traverse a binary tree in inorder, preorder, and postorder.				(L3, CO4)	6
B)	Construct the final binary search tree (BST) by inserting the following numbers in the given order: 40, 20, 50, 10, 30, 60, 25, 35, 55, 70, 5, 15, 65, 75, 45 1. Draw the final binary search tree. 2. Write the pre-order traversal of the tree.				(L3, CO4)	6
C)	Describe the adjacency matrix representation of graphs with an example.				(L2, CO4)	6
Q. 6	Solve Any Two of the following.					12
A)	Implement a stack using an array. Write a program to perform push and pop operations.				(L3, CO2)	6
B)	Evaluate the postfix expression 15 7 1 1+ - / 3 * 2 1 1+ + - where 15, 7, 1, and 3 are operands. Show step-by-step using a stack.				(L3, CO2)	6
C)	Write an algorithm to evaluate the postfix expression using stack				(L3, CO2)	6
*** End ***						